JAMIE MONTWé PRESENTS

PLANT PICTURE FRAME

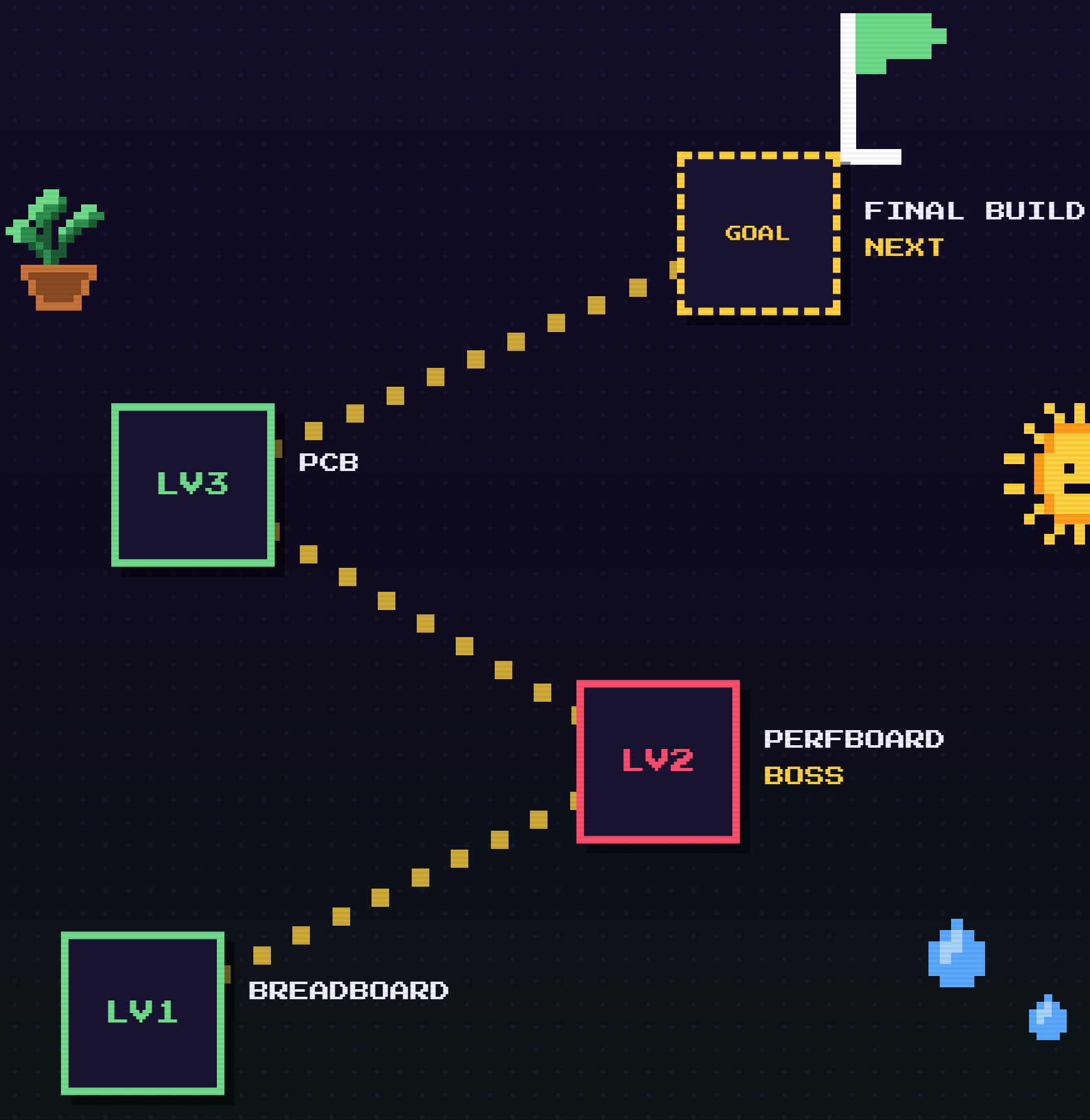


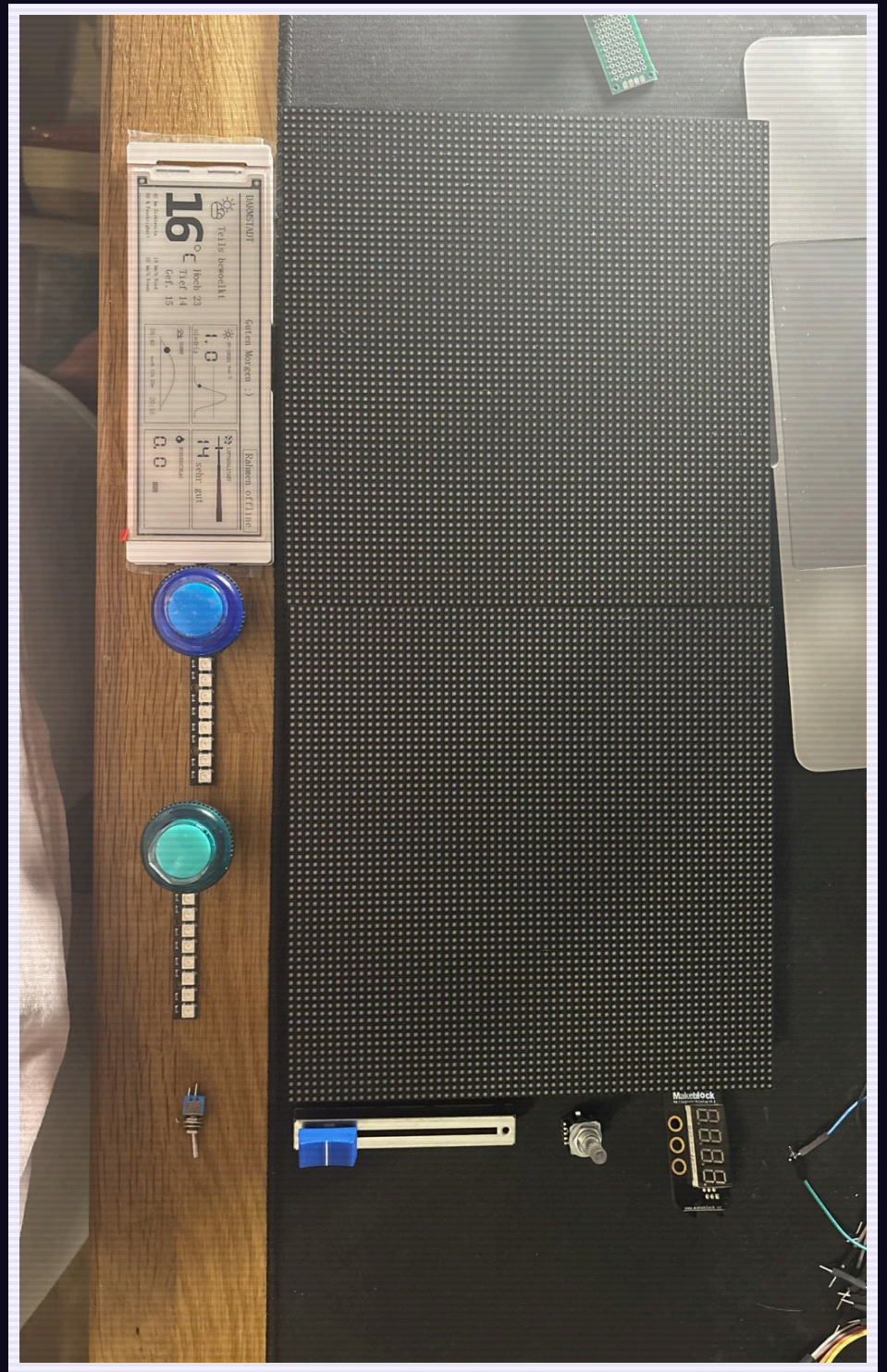
▶ PRESS START

© 2026 · 1 PLAYER · KICAD · ESP32 · ARDUINO · NX

The route so far

A digital plant on two LED matrices that reacts to real light, temperature and humidity. You water and mist it with buttons on the frame.





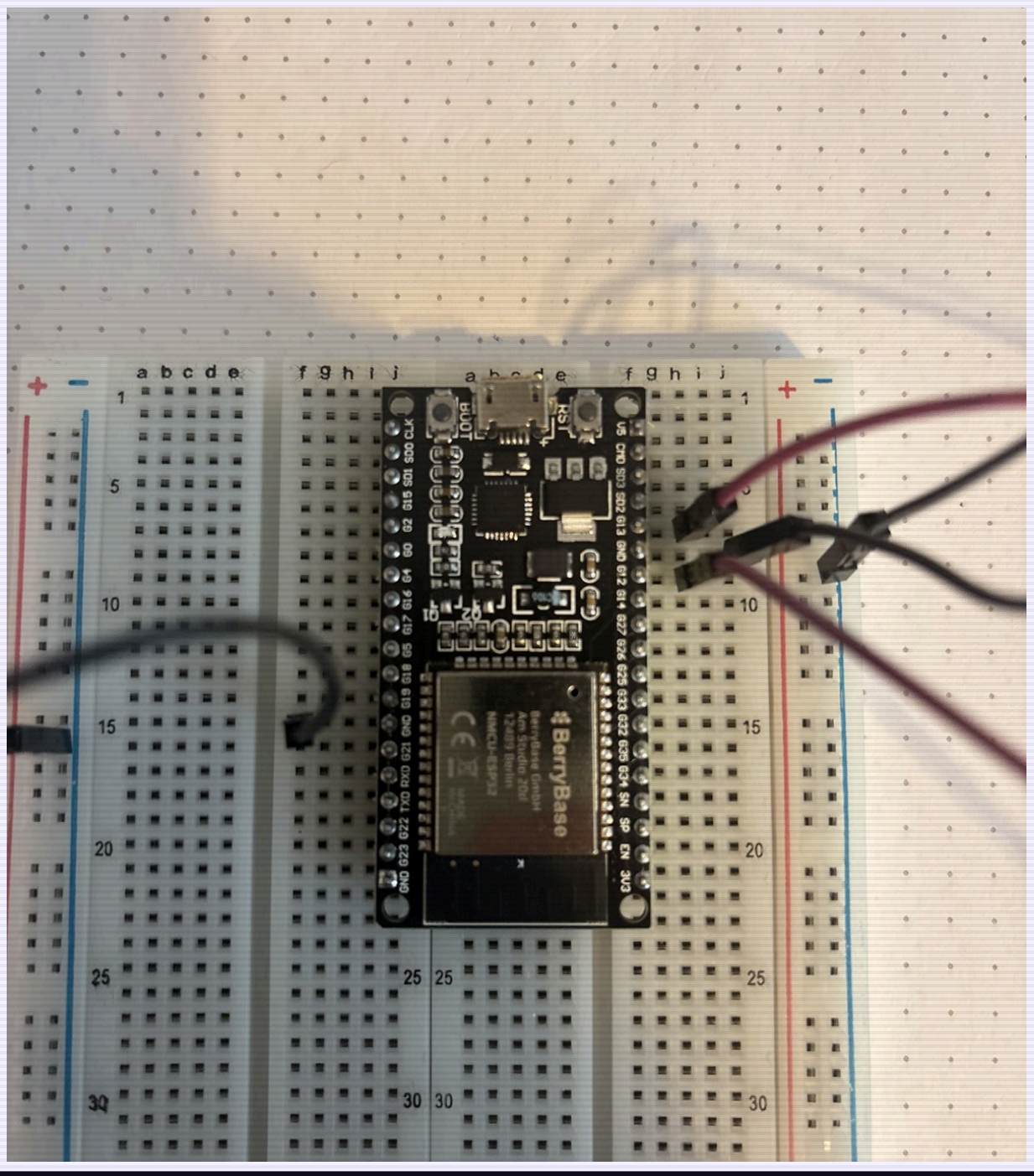
Items collected

Everything laid out before a single wire was connected.

- ESP32 x1
- LED MATRIX x2
- E-PAPER x1
- ARCADE BTN x2
- ENCODER x1
- SLIDER x1
- TM1637 x1
- SENSORS x2

TIP

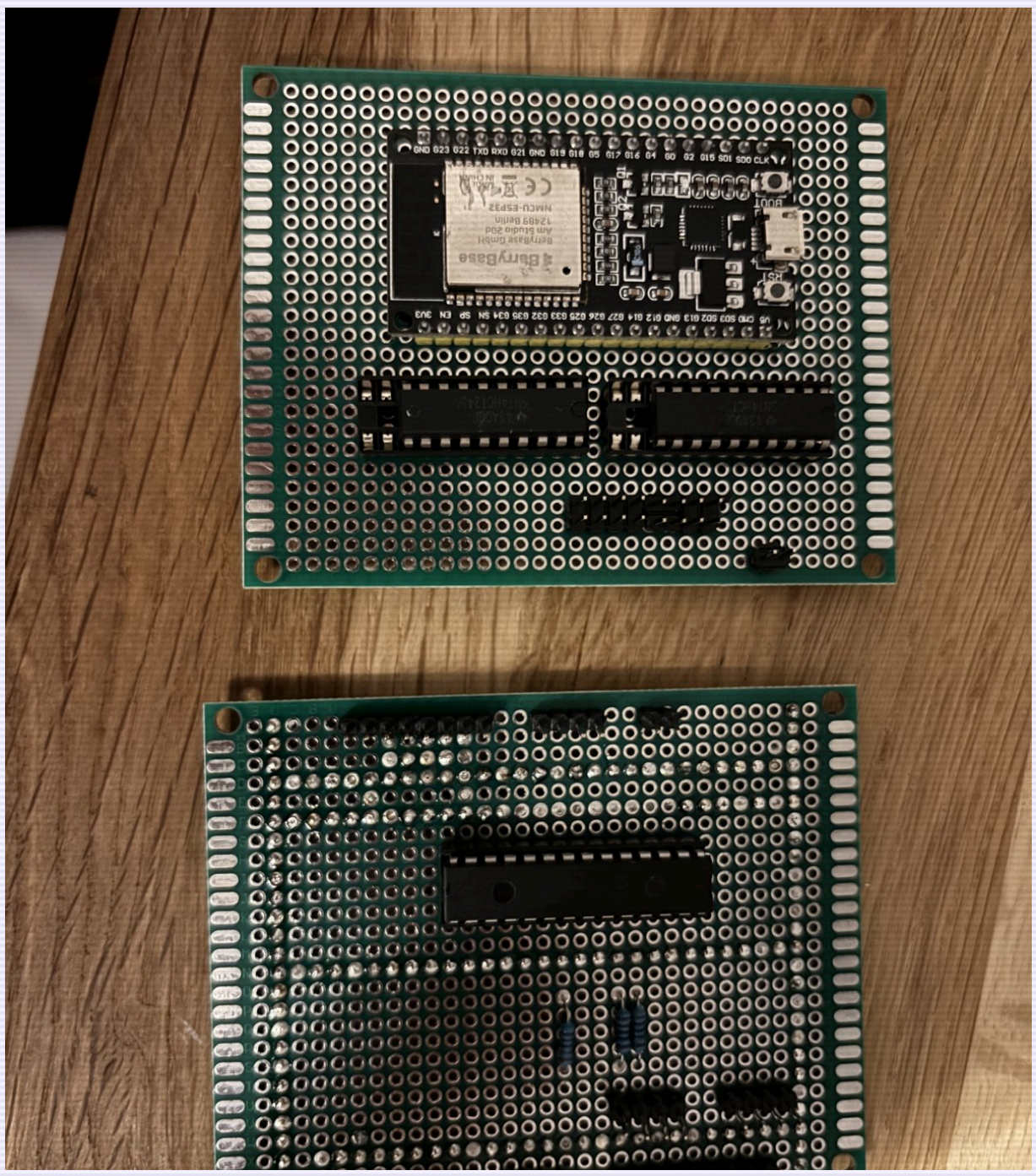
Light comes from a VEML7700, temperature and humidity from an SHT31. Both talk to the ESP32 over I²C.



LEVEL 1 · BREADBOARD

First signs of life. Arduino and C++ were new to me, so every part got its own small test sketch first.

XP 25%

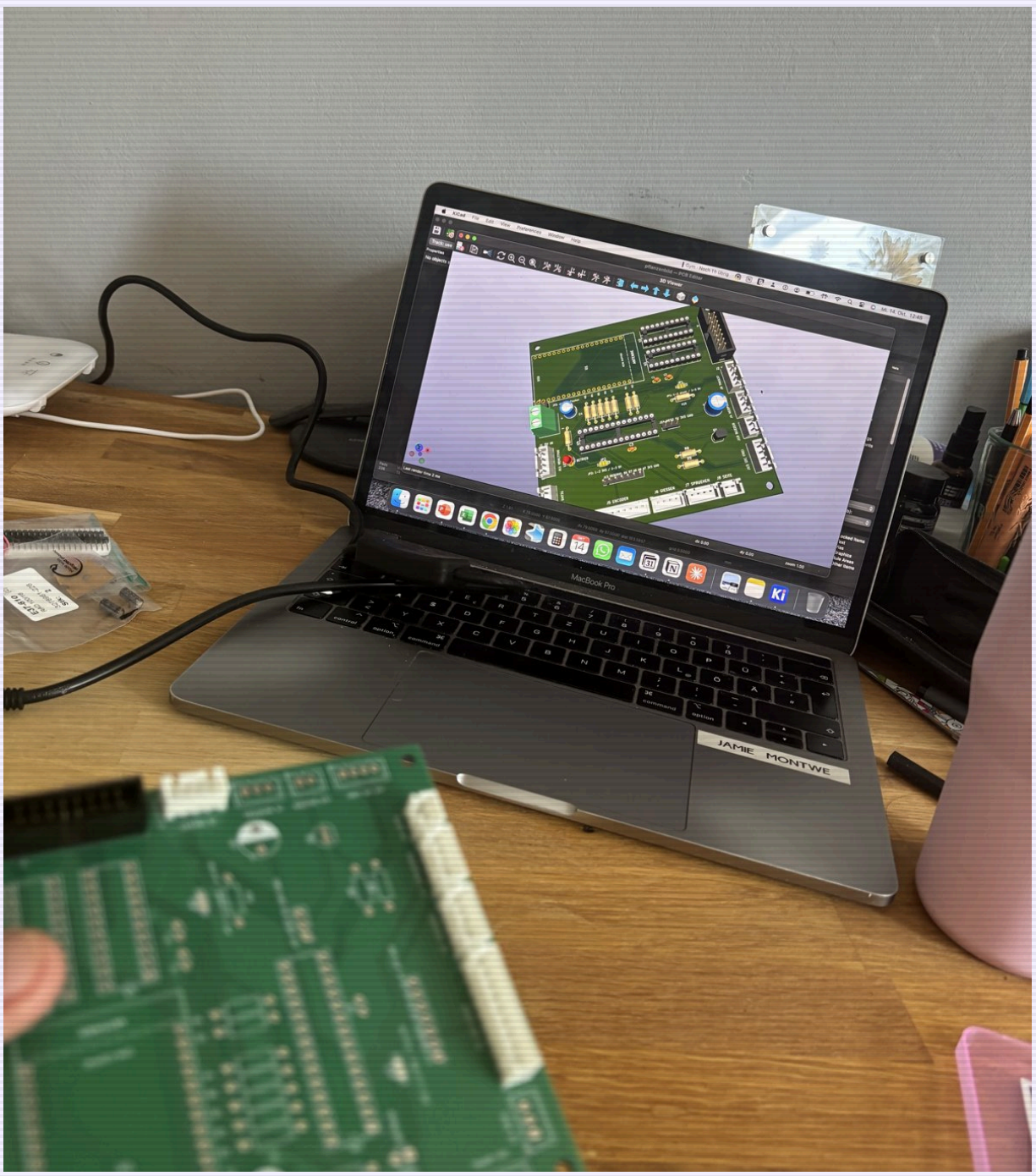


BOSS: SOLDERED-IN ESP32 DEFEATED

LEVEL 2 · PERFBOARD

I skipped the breadboard tests and soldered the ESP32 straight in. Getting it out meant desoldering all 38 pins and fitting a new header. On the PCB it sits in a socket.

XP 50%



LEVEL 3 · PCB

From KiCad model to a real board. 4 revisions, then manufactured by JLCPCB.

XP 75%



LEVEL 3 · CLEARED

Hand soldered and labelled. ESP32, MCP23017 port expander, sensors over I²C and a separate 5 V feed for the panels.

XP 100%

PLAYER STATS

8 / 8

PCB REVISIONS

4

BUDGET

~€200

PINS DESOLDERED

38

PLANT HP



ELECTRONICS

- KiCad PCB
- ESP32
- I²C sensors
- E-paper

SOFTWARE

- C++
- Arduino
- No extra sensor libraries

MECHANICAL

- Siemens NX
- Prusa MK3S print

ACHIEVEMENT UNLOCKED

Labels and datasheets don't always match the real hardware. Measure it yourself.

CONTINUE?

▸ YES NO

Next level: final assembly in the frame.

JAMIE MONTWÉ

linkedin.com/in/jamie-montwe